

Jialun Cao

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Citations: 699, H-index: 15, i10-index: 18.
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RESEARCH INTERESTS

My research focuses on automating software testing and program verification using advanced AI techniques.

- **Software Engineering (SE), Artificial Intelligence (AI), Formal Methods (FM)**
- **Research Topics:** AI for SE (e.g., LLM for Code), SE for AI (e.g., AI system testing)
- **Keywords:** [AI Testing](#), [AI for SE](#), [AI for FM](#), [LLM Benchmarking](#)

WORKING EXPERIENCE

- **Research Assistant Professor** Aug 2024 – Now
 - The Hong Kong University of Science and Technology
 - Department of Computer Science and Engineering
- **Mentee** Dec – 2024 – Nov 2025
 - Asia Pacific Women in Leadership (APWiL) Mentoring Program
 - Mentor: Prof. Paola Ricarte Quijano at **Harvard University and Tecnológico de Monterrey**
- **Visiting Scholar** Oct 2024 – Jan 2025
 - University of Stuttgart
 - Hosting professor: Prof. Michael Pradel (**ACM Distinguished Member**)
 - Collaboration topic: LLM for test generation
- **Postdoctoral Fellow** Apr – July 2024
 - The Hong Kong University of Science and Technology
 - Hosting professor: Prof. Shing-Chi Cheung
- **Visiting Scholar** Mar – Apr 2024
 - **The Chinese University of Hong Kong (Shenzhen)**
 - Hosting professor: Prof. Pinjia He
 - Collaboration topic: LLM benchmarking
- **Intern** Jul– Sep 2023
 - **Fermat Lab**, Huawei Hong Kong
 - Hosting professor: Prof. Shengchao Qin
- **Intern** Sep 2022 – Jan 2023
 - **Trusted Software Engineering and Open-Source Laboratory**, Huawei Hong Kong
 - Hosting professor: Prof. Xin Xia (**ACM Distinguished Member**)
- **Intern** Sep 2022 – Jan 2023
 - **Theory Lab**, Central Research Institute, 2012 Labs, Huawei Hong Kong
 - Hosting professor: Dr. Rui Zhang

ACADEMIC BACKGROUND

- **Ph.D., The Hong Kong University of Science and Technology** Sep 2019 – Mar 2024
 - Department of Computer Science and Engineering
 - **Supervisor:** Prof. Shing-Chi Cheung, **IEEE Fellow, ACM Distinguished Member**
 - **Research areas:** AI Testing, Software Testing, LLM for SE
 - **Thesis:** Towards Automatic Testing and Fault Localization in Natural Language Processing Systems
- **M.S., University of Chinese Academy of Sciences** Sep 2016 – Sep 2019
 - State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences
 - **Supervisor:** Prof. Yongjian Li
 - **Research areas:** Formal Verification, Parameterized Model Checking
- **B.S., Shandong University** Sep 2012 - Jun 2016
 - School of Mechanical, Electrical & Information Engineering, Computer Science Department

HONORS AND AWARDS

- 2025 **ACM SIGSOFT Outstanding Doctoral Dissertation Award**
- 2024 **ACM Distinguished paper award (ASE)**
- 2019 – 2021 Huawei Fellowship Scholarship
- 2023 – 2024 Hong Kong Postgraduate Scholarship
- 2017 **National Scholarship** awarded by Minister of Education (**Postgraduate, Rank 2/106, Top 1%**)
- 2014 **National Scholarship** awarded by Minister of Education (**Undergraduate, Rank 1/52, Top 1**)

FUNDING EXPERIENCE

- **Co-PI**
 - “A Multi-faceted Framework for Automated Pseudocode-to-Code Translation using Large Language Models”
 - 2025/26 General Research Fund project funded by Research Grants Council of Hong Kong.
 - [REDACTED]
- **Key contributor**
 - “Large language models driven Test Oracle Generation” 2023.12.21 – 2025.05.06
 - [REDACTED]
 - “From requirements to formal modeling verification projects based on large language models” 2024.01.19 – 2025.04.23
 - [REDACTED]
 - “Large language models aided program abstraction and model checking” 2024.10.30 – Now
 - [REDACTED]

PROFESSIONAL SERVICES (Selected)

- **Award Committee Member**
 - ACM SIGSOFT Outstanding Research Award 2025
 - ACM SIGSOFT Distinguished Service Award 2025
 - ACM SIGSOFT Influential Educator Award 2025
- **Local Chair**
 - Frontiers Forum on AI and Formal Methods 2025, Hong Kong ([Website](#))
- **Session Chair**
 - ACM International Conference on Automated Software Engineering (ASE 2024)
 - ACM International Conference on the Foundations of Software Engineering 2025 (FSE 2025)
 - International Conference on Internetwork 2025
- **Conference Program Committee Member**
 - IEEE/ACM International Conference on Software Engineering (ICSE) 2025 - 2026
 - ACM International Conference on the Foundations of Software Engineering (FSE) 2025 - 2026
 - IEEE/ACM International Conference on Automated Software Engineering (ASE) 2025
 - International Conference on Internetwork 2024 - 2025
 - IEEE International Symposium on Software Reliability Engineering (ISSRE) 2024
 - IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER 2025)
 - Asia-Pacific Software Engineering Conference (ASPEC) 2024
- **Journal Reviewer**
 - ACM Transactions on Software Engineering and Methodology (TOSEM)
 - IEEE Transactions on Software Engineering (TSE)
 - Empirical Software Engineering (EMSE)
 - ACM Transactions on Knowledge Discovery from Data (TKDD)
 - IEEE Transactions on Mobile Computing (TMC)

TEACHING EXPERIENCE

- **Instructor**
 - HKUST COMP1021 Introduction to Computer Science. Spring 2025.
 - **Enroll: 110 undergraduate students**
- **Guest Instructor**
 - Nankai University 2025 International Frontier Courses. Summer 2025. (Enroll: **228 students**)
- **Teaching Assistant**
 - HKUST COMP1021 Introduction to Computer Science. Fall 2023.
 - HKUST COMP3021 Java Programming, Fall 2020.
 - HKUST COMP3021 Java Programming, Spring 2020.

MENTORING EXPERIENCE

- Xiao Chen, current **PhD student** at HKUST
- Jingyi Chen, current **PhD student** at HKUST
- Dongze Li, current **PhD student** at HKUST
- Congying Xu, current **PhD student** at HKUST
- Songqiang Chen, current **PhD student** at HKUST
- Hau Ching Lo, current **MSc student** at HKUST
- Zhiyong Chen, current **MSc student** at Nanjing University
- Qiming Zhu, current **MSc student** at Chinese Academy of Science
- Ruiyang Xu, current **Msc student** at Chinese Academy of Science
- Ujaan Das, previous **undergraduate student** at HKUST, now a Msc student in University of Warwick

ADMINISTRATIVE SERVICES

- **Final-Year-Project**
 - Second reader: Year 2024 – 2025
 - Co-supervisor: Year 2021 – 2022
- **Hong Kong Joint University Program Admissions (JUPAS)**
 - Interviewer: Year 2025
- **Joint Entrance Examination (JEE) Recruitment in Mainland China**
 - Speaker: Year 2025

INVITED TALKS (SELECTED)

- **From Informal to Formal: Incorporating and Evaluating LLMs on Natural Language Requirements to Verifiable Formal Proofs.**
 - In Formal Methods Committee Strategic Seminar
 - Invited by China Computer Federation (CCF) Formal Methods Committee, May 2025
- **Exploring Code Generation and Reasoning Capabilities of LLMs.**
 - In Applied Mathematics Seminar at **Peking University**
 - Invited by **Prof. Meng Sun**, Apr 2025
- **From Benchmarks to Practice: A Study on the Code Capabilities of Large Language Models.**
 - In the Next Era of AI-assisted R&D Seminar, Hong Kong
 - Invited by Hong Kong Science Park, Jan 2025
- **Is Large Language Model a Rescue for Code Generation and Code Reasoning?**
 - In **University of Stuttgart**
 - Invited by Prof. Michael Pradel, Nov 2024
- **Can AI be a Panacea? Exploring Automatic Testing and Verification for Software Programs.**
 - In the Software Systems Engineering Group at **University College London (UCL)**
 - Invited by **Prof. Federica Sarro**, Oct 2024
- **Trusted Architecture of Intelligent CPS Systems.**
 - In Micro-Forum of Intelligent Software Development at **Fudan University**
 - Invited by **Prof. Xin Peng**, Aug 2024
- **From Requirement to Formal Specification and Model Generation via Large Language Models.**
 - In Formal Methods Committee Strategic Seminar
 - Invited by China Computer Federation (CCF) Formal Methods Committee, May 2024
- **Crafting Future: A Dancer's Leap into Computer Science.**
 - In the **1st Forum for Women Scholars in Software Engineering** in CCF ChinaSoft.
 - Invited by Prof. Ying Wang from Northeastern University, Dec 2023
- **A Study on the Problem of Data Contamination in the Era of Large Language Models.**
 - In New Paradigm of Software Engineering under AIGC in CCF ChinaSoft.
 - Invited by **Prof. Xin Xia** from Zhejiang University, Dec 2023

SELECTED PUBLICATIONS

1. **[ACL'25a (Main)] Jialun Cao**, Yaojie Lu, Meiziniu Li, Haoyang Ma, Haokun Li, Mengda He, Cheng Wen, Le Sun, Hongyu Zhang, Shengchao Qin, Shing-Chi Cheung, and Cong Tian. 2025. From Informal to Formal – Incorporating and Evaluating LLMs on Natural Language Requirements to Verifiable Formal Proofs. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, pages 26984–27003. **[38k social media views, 2k+ Model Downloads, 2k+ Dataset Downloads]**
2. **[ASE'24a] Jialun Cao**, Zhiyong Chen, Jiarong Wu, Shing-Chi Cheung, and Chang Xu. "Javabench: A benchmark of object-oriented code generation for evaluating large language models." In Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering, pp. 870-882. 2024.
3. **[ICSE'22] Jialun Cao**, Meiziniu Li, Xiao Chen, Ming Wen, Yongqiang Tian, Bo Wu, and Shing-Chi Cheung. "DeepFD: Automated fault diagnosis and localization for deep learning programs." In Proceedings of the 44th international conference on software engineering, pp. 573-585. 2022.
4. **[TOSEM'22] Jialun Cao**, Meiziniu Li, Yeting Li, Ming Wen, Shing-Chi Cheung, and Haiming Chen. "SemMT: a semantic-based testing approach for machine translation systems." ACM Transactions on Software Engineering and Methodology (TOSEM) 31, no. 2 (2022): 1-36.
5. **[CAV'24] Cheng Wen, Jialun Cao (Corresponding)**, Jie Su, Zhiwu Xu, Shengchao Qin (Corresponding), Mengda He, Haokun Li, Shing-Chi Cheung, and Cong Tian. "Enchanting program specification synthesis by large language models using static analysis and program verification." In International Conference on Computer Aided Verification, pp. 302-328. Cham: Springer Nature Switzerland, 2024.

FULL PUBLICATION LIST (Chronological Order)

1. **[ACL'25a (Main)] Jialun Cao**, Yaojie Lu, Meiziniu Li, Haoyang Ma, Haokun Li, Mengda He, Cheng Wen, Le Sun, Hongyu Zhang, Shengchao Qin, Shing-Chi Cheung, and Cong Tian. 2025. From Informal to Formal – Incorporating and Evaluating LLMs on Natural Language Requirements to Verifiable Formal Proofs. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, pages 26984–27003. **[38k social media Views, 2k+ Model Downloads, 2k+ Dataset Downloads]**
2. **[JASE'25] Jialun Cao**, Meiziniu Li, Ming Wen, and Shing-chi Cheung. "A study on prompt design, advantages and limitations of ChatGPT for deep learning program repair." Automated Software Engineering 32, no. 1 (2025): 1-29.
3. **[ACM SIGSOFT Software Engineering Notes'25] Jialun Cao**. "Towards Automatic Testing and Fault Localization in Natural Language Processing Systems." ACM SIGSOFT Software Engineering Notes 50, no. 2 (2025): 11-16.
4. **[Internetwork'25] Jialun Cao**, Songqiang Chen, Wuqi Zhang, Hau Ching Lo, and Shing-Chi Cheung. "CodeCleaner: Elevating Standards with A Robust Data Contamination Mitigation Toolkit." In 16th International Conference on Internetwork June 20–22, 2025.
5. **[ACL'25b] Ruiyang Xu***, **Jialun Cao *(Co-1st)**, Yaojie Lu, Ming Wen, Hongyu Lin, Xianpei Han, Ben He, Shing-Chi Cheung, and Le Sun. 2025. CRUXEVAL-X: A Benchmark for Multilingual Code Reasoning, Understanding and Execution. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, pages 23762–23779.
6. **[AAAI'25a] Qiming Zhu***, **Jialun Cao *(Co-1st)**, Yaojie Lu, Hongyu Lin, Xianpei Han, Le Sun, and Shing-Chi Cheung. "Domaineval: An auto-constructed benchmark for multi-domain code generation." In Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 24, pp. 26148-26156. 2025.
7. **[FSE'25] Xiao Chen**, Hengcheng Zhu, **Jialun Cao (Corresponding)**, Ming Wen, and Shing-Chi Cheung (Corresponding). "SemBIC: Semantic-Aware Identification of Bug-Inducing Commits." Proceedings of the ACM on Software Engineering, 2025. pp.1363-1385.
8. **[ASE'25] Xingchu Chen**, Chengwei Liu, **Jialun Cao**, Yang Xiao, Xinyue Cai, Yeting Li, Jingyi Shi, Tianqi Sun, Haiming Chen, Wei Huo. "Vulnerability-Affected Versions Identification: How Far Are We?" In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (2025).
9. **[TOSEM'25] Meiziniu Li**, Dongze Li, Jianmeng Liu, **Jialun Cao**, Yongqiang Tian, and Shing-Chi Cheung. "Enhancing Differential Testing with LLMs For Testing Deep Learning Libraries." ACM Transactions on Software Engineering and Methodology (2025).

10. **[AAAI'25b]** Mengyang Wu, Yuzhi Zhao, **Jialun Cao**, Mingjie Xu, Zhongming Jiang, Xuehui Wang, Qinbin Li, Guangneng Hu, Shengchao Qin, and Chi-Wing Fu. "ICM-Assistant: Instruction-tuning Multimodal Large Language Models for Rule-based Explainable Image Content Moderation." In Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 8, pp. 8413-8422. 2025.
11. **[TDSC'25]** Hongjie Gu, Daojing He, **Jialun Cao**, Gaolei Li, Kim-Kwang Raymond Choo. "HGAFa: Heterogeneous Graph Attention-based Featureless Aggregation for IoC Joint Identification". In IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
12. **[ASE'24a]** **Jialun Cao**, Zhiyong Chen, Jiarong Wu, Shing-Chi Cheung, and Chang Xu. "Javabench: A benchmark of object-oriented code generation for evaluating large language models." In Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering, pp. 870-882. 2024.
13. **[ASE'24b Distinguished Paper Award]** Zongze Jiang, Ming Wen, **Jialun Cao**, Xuanhua Shi, and Hai Jin. "Towards understanding the effectiveness of large language models on directed test input generation." In Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering, pp. 1408-1420. 2024.
14. **[ASE'24c]** Congying Xu, Songqiang Chen, Jiarong Wu, Shing-Chi Cheung (Corresponding), Valerio Terragni, Hengcheng Zhu, and **Jialun Cao (Corresponding)**. "MR-Adopt: Automatic deduction of input transformation function for metamorphic testing." In Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering, pp. 557-569. 2024.
15. **[CAV'24]** Cheng Wen, **Jialun Cao (Corresponding)**, Jie Su, Zhiwu Xu, Shengchao Qin (Corresponding), Mengda He, Haokun Li, Shing-Chi Cheung, and Cong Tian. "Enchanting program specification synthesis by large language models using static analysis and program verification." In International Conference on Computer Aided Verification, pp. 302-328. Cham: Springer Nature Switzerland, 2024.
16. **[APSEC'24]** Yang Bo, Jiawei Hu, and **Jialun Cao (Corresponding)**. "SDEFL: A Lightweight Fault Detection and Localization Method for Deep Neural Networks." In 2024 31st Asia-Pacific Software Engineering Conference (APSEC), pp. 321-332. IEEE, 2024.
17. **[TASE'24]** Kunpeng Jian, Yanyan Zou, Yeting Li, **Jialun Cao**, Menghao Li, Jian Sun, Jingyi Shi, and Wei Huo. "Fuzzing for Stateful Protocol Implementations: Are We There Yet?" In International Symposium on Theoretical Aspects of Software Engineering, pp. 186-204. Cham: Springer Nature Switzerland, 2024.
18. **[FSE'23a]** **Jialun Cao**, Yaojie Lu, Ming Wen, and Shing-Chi Cheung. "Testing coreference resolution systems without labeled test sets." In Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, pp. 107-119. 2023.
19. **[TOSEM'23]** Meiziniu Li, **Jialun Cao**, Yongqiang Tian, Tsz On Li, Ming Wen, and Shing-Chi Cheung. "Comet: Coverage-guided model generation for deep learning library testing." ACM Transactions on Software Engineering and Methodology 32, no. 5 (2023): 1-34.
20. **[FSE'23b]** Xiaohu Du, Xiao Chen, **Jialun Cao**, Ming Wen, Shing-Chi Cheung, and Hai Jin. "Understanding the bug characteristics and fix strategies of federated learning systems." In Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, pp. 1358-1370. 2023.
21. **[ICSE'22]** **Jialun Cao**, Meiziniu Li, Xiao Chen, Ming Wen, Yongqiang Tian, Bo Wu, and Shing-Chi Cheung. "DeepFD: Automated fault diagnosis and localization for deep learning programs." In Proceedings of the 44th international conference on software engineering, pp. 573-585. 2022.
22. **[TOSEM'22]** **Jialun Cao**, Meiziniu Li, Yeting Li, Ming Wen, Shing-Chi Cheung, and Haiming Chen. "SemMT: a semantic-based testing approach for machine translation systems." ACM Transactions on Software Engineering and Methodology (TOSEM) 31, no. 2 (2022): 1-36.
23. **[USENIX Security'22]** Yeting Li, Yecheng Sun, Zhiwu Xu, **Jialun Cao**, Yuekang Li, Rongchen Li, Haiming Chen, Shing-Chi Cheung, Yang Liu, and Yang Xiao. "RegexScalpel: Regular Expression Denial of Service (ReDoS) Defense by Localize-and-Fix." In 31st USENIX Security Symposium (USENIX Security 22), pp. 4183-4200. 2022.
24. **[USENIX Security'21]** Yeting Li, Zixuan Chen, **Jialun Cao**, Zhiwu Xu, Qiancheng Peng, Haiming Chen, Liyuan Chen, and Shing-Chi Cheung. "ReDoSHunter: A combined static and dynamic approach for regular expression DoS detection." In 30th USENIX Security Symposium, pp. 3847-3864. 2021.

25. **[ICSE'21]** Yeting Li, Shuaimin Li, Zhiwu Xu, **Jialun Cao**, Zixuan Chen, Yun Hu, Haiming Chen, and Shing-Chi Cheung. "TransRegex: multi-modal regular expression synthesis by generate-and-repair." In 2021 IEEE/ACM 43rd International Conference on Software Engineering (ICSE), pp. 1210-1222. IEEE, 2021.
26. **[ASE'20]** Yeting Li, Zhiwu Xu, **Jialun Cao**, Haiming Chen, Tingjian Ge, Shing-Chi Cheung, and Haoren Zhao. "Flashregex: Deducing anti-redos regexes from examples." In Proceedings of the 35th IEEE/ACM International Conference on Automated Software Engineering, pp. 659-671. 2020.
27. **[ICDE'20]** Yeting Li, **Jialun Cao**, Haiming Chen, Tingjian Ge, Zhiwu Xu, and Qiancheng Peng. "FlashSchema: achieving high quality XML schemas with powerful inference algorithms and large-scale schema data." In 2020 IEEE 36th International Conference on Data Engineering (ICDE), pp. 1962-1965. IEEE, 2020.
28. **[DASFFA'19]** Yeting Li, Xiaolan Zhang, **Jialun Cao**, Haiming Chen, and Chong Gao. "Learning k-occurrence regular expressions with interleaving." In International Conference on Database Systems for Advanced Applications, pp. 70-85. Cham: Springer International Publishing, 2019.
29. **[ICCD'19]** Yongjian Li, **Jialun Cao (1st Student Author)**, and Jun Pang. "A learning-based framework for automatic parameterized verification." In 2019 IEEE 37th International Conference on Computer Design (ICCD), pp. 450-459. IEEE, 2019.
30. **[ASE'18 Demo]** **Jialun Cao**, Yongjian Li, and Jun Pang. "L-CMP: An automatic learning-based parameterized verification tool." In Proceedings of the 33rd ACM/IEEE International Conference on Automated Software Engineering, pp. 892-895. 2018.
31. **[QRS'18]** Yongjian Li, **Jialun Cao (1st Student Author)**, and Kaiqiang Duan. "An automatic parameterized verification of FLASH cache coherence protocol." In 2018 IEEE International Conference on Software Quality, Reliability and Security (QRS), pp. 47-58.

WORK IN PROGRESS

32. **[Preprint'25a]** **Jialun Cao**, Yuk-Kit Chan, Zixuan Ling, Wenxuan Wang, Shuqing Li, Mingwei Liu, Ruixi Qiao et al. "How Should We Build a Benchmark? Revisiting 274 Code-Related Benchmarks for LLMs." arXiv preprint arXiv:2501.10711.
33. **[Preprint'25b]** **Jialun Cao**, Wuqi Zhang, and Shing-Chi Cheung. "Concerned with data contamination? assessing countermeasures in code language model." arXiv preprint arXiv:2403.16898.
34. **[Under Review'25a]** **Jialun Cao**, Haoran Yan, Haoyu Wang, Ujaan Das, Ming Wen, Michael Pradel. "STARS: Static Analysis-guided Assertion Synthesis Using LLMs. "
35. **[Under Review'25b]** Xiao He, Ru Chen, **Jialun Cao (Corresponding)**. "From What to How: Bridging User Requirements with Software Development Using Large Language Models."
36. **[Preprint'25c]** Jiarong Wu, Songqiang Chen, **Jialun Cao (Corresponding)**, Hau Ching Lo, and Shing-Chi Cheung (Corresponding). "Isolating language-coding from problem-solving: Benchmarking LLMs with pseudoeval." arXiv preprint arXiv:2502.19149.
37. **[Preprint'25d]** Jingyi Chen, Songqiang Chen, **Jialun Cao (Corresponding)**, Jiasi Shen (Corresponding), and Shing-Chi Cheung. "When LLMs Meet API Documentation: Can Retrieval Augmentation Aid Code Generation Just as It Helps Developers?." arXiv preprint arXiv:2503.15231.
38. **[Preprint'25e]** Dongze Li, Songqiang Chen, **Jialun Cao (Corresponding)**, and Shing-Chi Cheung (Corresponding). "What Builds Effective In-Context Examples for Code Generation?" arXiv preprint arXiv:2508.06414.
39. **[Preprint'25f]** Ruiyang Xu*, **Jialun Cao*(Co-1st)**, Mingyuan Wu, Wenliang Zhong, Yaojie Lu, Ben He, Xianpei Han, Shing-Chi Cheung, and Le Sun. "EmbedAgent: Benchmarking Large Language Models in Embedded System Development." arXiv preprint arXiv:2506.11003.
40. **[Under Review'25d]** Zhiyong Chen*, **Jialun Cao* (Co-1st)**, Jiarong Wu, Chang Xu, Shing-Chi Cheung. "Can Large Language Models Model Programs Formally? "
41. **[Preprint'25g]** Xiaolei Li, **Jialun Cao**, Yepang Liu, Shing-Chi Cheung, and Hailong Wang. "Enhancing LLM-Based GUI Test Migration via Key Intention Inference and Visual Analysis." arXiv preprint arXiv:2504.02357.

42. **[Preprint'25h]** Qiming Zhu, **Jialun Cao**, Xuanang Chen, Yaojie Lu, Hongyu Lin, Xianpei Han, Le Sun, and Shing-Chi Cheung. "Across Programming Language Silos: A Study on Cross-Lingual Retrieval-augmented Code Generation." arXiv preprint arXiv:2506.03535.
43. **[Under Review'25e]** Junyi Wang, **Jialun Cao**, Zhongxin Liu. "iCoRe: An Iterative Correlation-Aware Retriever for Bug Reproduction Test Generation."
44. **[Preprint'25i]** Le Deng, Zhonghao Jiang, **Jialun Cao**, Michael Pradel, Zhongxin Liu. "NoCode-bench: A Benchmark for Evaluating Natural Language-Driven Feature Addition." arXiv preprint arXiv:2507.18130.
45. **[Preprint'25j]** Dekun Dai, MingWei Liu, Anji Li, **Jialun Cao**, Yanlin Wang, Chong Wang, Xin Peng, and Zibin Zheng. "FeedbackEval: A Benchmark for Evaluating Large Language Models in Feedback-Driven Code Repair Tasks." arXiv preprint arXiv:2504.06939.
46. **[Preprint'25k]** Songqiang Chen, Congying Xu, Jingyi Chen, **Jialun Cao (Corresponding)**, Jiarong Wu, Shing-Chi Cheung (Corresponding). "On Effective Semantic Translation for Code: A Study Based on Pseudocode." arXiv preprint arXiv: 2510.00920.